

Name:Naveena C J

USN: 1AM22CD061

e-mail:Naveenacj99@gmail.com

lab 9:

```
data class Item(val name: String, val category: String, val quantity: Int)
```

```
fun printInventory(items: List<Item>) {  
    for (item in items) {  
        println("Name: ${item.name} | Category: ${item.category} | Qty: ${item.quantity}")  
    }  
    if (items.isEmpty()) {  
        println("No items found.")  
    }  
    println()  
}
```

```
fun <T> filterList(list: List<T>, condition: (T) -> Boolean): List<T> {  
    return list.filter(condition)  
}
```

```
fun main() {  
    val inventory = listOf(  
        Item("Laptop", "Electronics", 12),  
        Item("Keyboard", "Electronics", 5),  
        Item("Chair", "Furniture", 20),  
        Item("Table", "Furniture", 8),  
        Item("Notebook", "Stationery", 50)
```

```
)
```

```
println("--- Full Inventory ---")
```

```
printInventory(inventory)
```

```
val lowStock = filterList(inventory) { it.quantity < 10 }
```

```
println("--- Low Stock (<10) ---")
```

```
printInventory(lowStock)
```

```
val electronics = filterList(inventory) { it.category == "Electronics" }
```

```
println("--- Electronics Category ---")
```

```
printInventory(electronics)
```

```
println("--- Total Quantity per Category ---")
```

```
val totals = inventory
```

```
    .groupBy { it.category }
```

```
    .mapValues { entry -> entry.value.sumOf { it.quantity } }
```

```
for ((category, total) in totals) {
```

```
    println("$category: $total")
```

```
}
```

```
}
```